WORKED SOLUTION

STEP I Specimen paper, Question 9

Pure • Modulus Inequalities

STEP I Specimen paper, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q9

The key idea

Both parts are about getting rid of a modulus sign by finding exactly where the expression inside changes sign or crosses a level. In (i), writing $u = x + 1$ turns the expression into $u - 2/u$, which is increasing on each side of $u = 0$, so the inequality just pins u between two roots of a quadratic on each branch. In (ii), writing $\cos x - \sin x$ as a single cosine shows where it changes sign, and we split the integral there.

Solution

Part (i)

We need $x \neq -1$ for the expression to be defined. Write $u = x + 1$, so $u \neq 0$ and

$$x + \frac{x-1}{x+1} = \frac{x^2+2x-1}{x+1} = \frac{(x+1)^2-2}{x+1} = u - \frac{2}{u}.$$

Call this $f(u)$. The inequality is $-2 < f(u) < 2$.

Ask yourself Why not just multiply through by $x + 1$?

Because $x + 1$ can be negative, and multiplying by a negative number reverses an inequality; doing it carelessly is the classic way to lose half the answer. Instead we use the shape of f. Its derivative is $f'(u) = 1 + 2/u^2 > 0$, so f is strictly increasing on $u < 0$ and, separately, on $u > 0$. On each branch it runs from $-\infty$ to $+\infty$: as $u \to 0^-$, $-2/u \to +\infty$; as $u \to 0^+$, $-2/u \to -\infty$; and $f(u) \sim u$ as $|u| \to \infty$. So on each branch f takes each of the values -2 and 2 exactly once, and $-2 < f < 2$ holds precisely between those two points.

For $u \neq 0$, multiplying by u gives two equivalent equations:

$$f(u) = 2 \iff u^2 - 2u - 2 = 0 \iff u = 1 \pm \sqrt{3},$$
$$f(u) = -2 \iff u^2 + 2u - 2 = 0 \iff u = -1 \pm \sqrt{3}.$$

Branch $u > 0$. The positive roots are $u = -1 + \sqrt{3}$ (where $f = -2$) and $u = 1 + \sqrt{3}$ (where $f = 2$). Since f is increasing, $-2 < f(u) < 2$ exactly when $-1 + \sqrt{3} < u < 1 + \sqrt{3}$, that is, $-2 + \sqrt{3} < x < \sqrt{3}$.

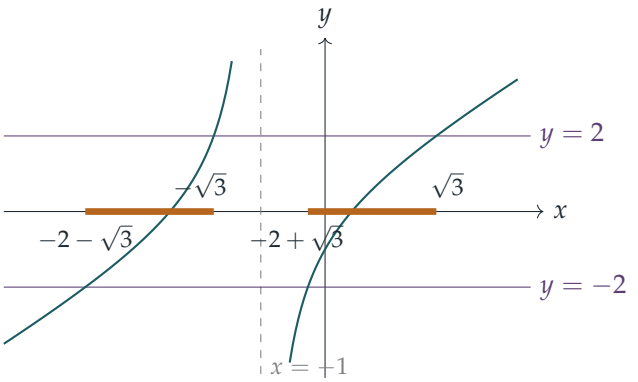
Branch $u < 0$. The negative roots are $u = -1 - \sqrt{3}$ (where $f = -2$) and $u = 1 - \sqrt{3}$ (where $f = 2$). Again by monotonicity, $-2 < f(u) < 2$ exactly when $-1 - \sqrt{3} < u < 1 - \sqrt{3}$, that is, $-2 - \sqrt{3} < x < -\sqrt{3}$.

Neither interval contains $x = -1$ (note $-\sqrt{3} < -1 < -2 + \sqrt{3}$), so the solution set is

$$\boxed{(-2 - \sqrt{3},\ -\sqrt{3}) \cup (-2 + \sqrt{3},\ \sqrt{3})}$$

with every endpoint of the form $a + b\sqrt{3}$ as required. The endpoints are excluded because there the modulus equals 2 exactly.

The graph of $y = x + (x-1)/(x+1)$ against the lines $y = \pm 2$ shows the two intervals (in amber) where the curve lies between the lines.



Part (ii)

Combine the two terms into one cosine:

$$\cos x - \sin x = \sqrt{2}\left(\tfrac{1}{\sqrt{2}}\cos x - \tfrac{1}{\sqrt{2}}\sin x\right) = \sqrt{2}\cos\left(x + \frac{\pi}{4}\right).$$

For $0 \le x \le \pi$ we have $\pi/4 \le x + \pi/4 \le 5\pi/4$, and cosine is positive on $[\pi/4, \pi/2)$ and negative on $(\pi/2, 5\pi/4]$. So $\cos x - \sin x \ge 0$ for $0 \le x \le \pi/4$ and ≤ 0 for $\pi/4 \le x \le \pi$. Splitting the integral at $\pi/4$, and using that an antiderivative of $\cos x - \sin x$ is $\sin x + \cos x$:

$$\begin{aligned}
\int_0^{\pi} |\cos x - \sin x|\,\mathrm{d}x &= \int_0^{\pi/4} (\cos x - \sin x)\,\mathrm{d}x - \int_{\pi/4}^{\pi} (\cos x - \sin x)\,\mathrm{d}x \\
&= \Big[\sin x + \cos x\Big]_0^{\pi/4} - \Big[\sin x + \cos x\Big]_{\pi/4}^{\pi} \\
&= (\sqrt{2} - 1) - (-1 - \sqrt{2}) \\
&= 2\sqrt{2}.
\end{aligned}$$

Checks

- Spot values in (i): at $x = 0$ the expression is -1, inside the modulus bound, and $0 \in (-2+\sqrt{3}, \sqrt{3})$; at $x = -2$ it is $-2+3 = 1$, and $-2 \in (-2-\sqrt{3}, -\sqrt{3})$; at $x = -1.5$ it is $-1.5+5 = 3.5$, outside, and indeed $-1.5 \in (-\sqrt{3}, -1)$ lies in neither interval.
- Testing the inequality on a fine grid and at 2×10^5 random points in $[-50, 50]$ agreed with the two intervals everywhere, and at all four endpoints the modulus equals 2.
- In (ii), Simpson's rule gives $2.828427\ldots$, matching $2\sqrt{2} \approx 2.828427$; the two pieces are $\sqrt{2}-1 \approx 0.4142$ and $\sqrt{2}+1 \approx 2.4142$. As a sanity check, the integrand is at most $\sqrt{2}$, so the integral is at most $\sqrt{2}\,\pi \approx 4.44$.

Common slips

- Multiplying both sides of $x + \frac{x-1}{x+1} < 2$ by $x+1$ without considering its sign, which loses (or wrongly includes) the interval to the left of $x = -1$.
- Solving only $f < 2$, or only $f > -2$: the modulus inequality needs both at once.
- In (ii), integrating $\cos x - \sin x$ straight across $[0, \pi]$, which gives -2 (a negative answer for the integral of a modulus is a red flag).

Beyond STEP

The trick in (i) is to replace an algebraic inequality by a statement about a monotonic function: if f is strictly increasing on an interval, then $a < f(u) < b$ holds exactly between the points where $f = a$ and $f = b$. Making this rigorous uses the intermediate value theorem, one of the first big results of a university analysis course, and inequalities with absolute values, such as $|x - a| < \varepsilon$, are the everyday language of that course. Abbott, *Understanding Analysis*, is a friendly place to see both.

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